AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 4
Contextual Applications of Differentiation

100-Question Master Examination

- Questions 1–90: Multiple Choice (Full Priority Ratings)
- Questions 91–100: Free Response with Graphs and Authentic AP Scoring Rubrics
- Complete Diagnostic Answer Key and Step-by-Step Solutions at the End

OFFICIAL AP CED ALIGNED · MIXED LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This paper contains 100 continuously numbered questions. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★	**Very Important**	High-frequency exam concepts: Motion (v, a, speed), Related Rates, Linearization, L'Hospital's Rule, and contextual interpretations with units.
★★★	**Mandatory**	Essential skills: Applied geometric rates, average vs instantaneous rates, concavity error bounds, and marginal analysis.
★★	**Important**	Foundational practice: Unit analysis of derivatives, basic tangent line equations, standard L'Hospital forms.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer.
- **Questions 91–100:** Free response, 9 points each, with graphs and contextual tables.
- **Difficulty Mix:** Easy 5 · Moderate 68 · Hard 17.
- **Calculator Policy:** No calculator required.
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 4.1 – 4.7)

4.1 Interpreting the Derivative in Context

4.2 Straight-Line Motion: Position, Velocity, Acceleration

4.3 Rates of Change in Applied Contexts

4.4 Introduction to Related Rates

4.5 Solving Related Rates Problems

4.6 Approximating Values Using Linearization

4.7 L'Hospital's Rule for Indeterminate Forms

> **Directions:** Attempt the entire paper before consulting the answer section. For free response questions, show all reasoning and state any theorem or rule by name.

QUESTION 01

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $P(t)$ is the population of a city t years after 2010, then $P'(4) = -250$ means

A. $P(4) = -250$

B. In 2014 the population is decreasing at 250 people per year

C. The city loses 250 people every 4 years

D. The population in 2014 is 250

QUESTION 02

EASY · MULTIPLE CHOICE

★★ IMPORTANT

A tank holds $V(t)$ liters of water after t minutes. The units of $V'(t)$ are

A. liters

B. minutes

C. liters per minute squared

D. liters per minute

QUESTION 03

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $C(x)$ is the cost in dollars of producing x items, then $C'(50) = 3.2$ means

A. Profit on 50 items is \$3.20

B. The 51st item costs about \$3.20 to produce

C. The cost of 50 items is \$3.20

D. Average cost is \$3.20

QUESTION 04

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The temperature T of a metal, in °C, t minutes after heating begins satisfies $T'(10) = 4$ and $T''(10) = -1$. Which is true at $t = 10$?

A. T is decreasing and the rate of increase is slowing

B. T is constant

C. T is increasing, but the rate of increase is slowing

D. T is increasing and the rate of increase is speeding up

QUESTION 05

EASY · MULTIPLE CHOICE

★★ IMPORTANT

If s(t) is position in meters and t is in seconds, the units of $s''(t)$ are

A. m/s^2

B. m/s

C. m

D. s/m

QUESTION 06

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle has velocity $v(t) = 3t^2 - 12t + 9$. It is at rest when

A. $t = 1$ only

B. $t = 3$ only

C. $t = 1$ and $t = 3$

D. $t = 0$

QUESTION 07

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $s(t) = t^3 - 6t^2 + 9t$, the acceleration at $t = 1$ is

A -6

B -12

C 0

D 6

QUESTION 08

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A particle moves with velocity $v(t) = t^2 - 4$. The speed at $t = 3$ is

A 1

B 9

C 5

D -5

QUESTION 09

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $v(t) < 0$ and $a(t) > 0$, then the particle is

A speeding up and moving right

B slowing down and moving left

C at rest

D speeding up and moving left

QUESTION 10

EASY · MULTIPLE CHOICE

★★★ MANDATORY

If $v(t) > 0$ and $a(t) > 0$, the particle is

A moving left

B slowing down

C speeding up in the positive direction

D changing direction

QUESTION 11

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$s(t) = t^2 - 4t + 3$. The particle changes direction when

A $t = 3$ only

B $t = 1$ only

C $t = 2$

D $t = 1$ and $t = 3$

QUESTION 12

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The average velocity on $[1, 4]$ for $s(t) = t^2$ is

A 8

B 5

C 3

D 7

QUESTION 13

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Instantaneous velocity at $t = a$ is defined by

A $s(a)$

B $\lim_{h \to 0} [s(a+h) - s(a)] / h$

C $[s(a+h) - s(a)] / h$

D $s''(a)$

QUESTION 14

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle has $a(t) = 6t$. If $v(0) = 4$, then $v(2) =$

A 8

B 16

C 12

D 24

QUESTION 15

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Displacement on $[0, 3]$ equals

A $\int_0^3 |v(t)|\, dt$

B $\int_0^3 v(t)\, dt$

C $\int_0^3 a(t)\, dt$

D $s(0)$

QUESTION 16

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Total distance traveled on [a, b] is

A |s(b) − s(a)|

B ∫_a^b v(t) dt

C max s − min s

D ∫_a^b |v(t)| dt

QUESTION 17

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Oil leaks from a tank so that $V'(t) = -2t$ gal/min. At $t = 3$, oil is leaking at a rate of

A 3 gallons per minute

B 2 gallons per minute

C 6 gallons per minute

D −6 gallons

QUESTION 18

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The radius of a circle increases at 2 cm/s. When $r = 5$ cm, $dA/dt =$

A 25π cm²/s

B 20π cm²/s

C 4π cm²/s

D 10π cm²/s

QUESTION 19

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The edge of a cube increases at 3 cm/s. When the edge is 2 cm, the volume increases at

A $8\ \text{cm}^3/\text{s}$

B $24\ \text{cm}^3/\text{s}$

C $12\ \text{cm}^3/\text{s}$

D $36\ \text{cm}^3/\text{s}$

QUESTION 20

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Water fills a cylindrical tank of radius 4 ft. If $dh/dt = 2$ ft/min, $dV/dt =$

A $4\pi\ \text{ft}^3/\text{min}$

B $32\pi\ \text{ft}^3/\text{min}$

C $16\pi\ \text{ft}^3/\text{min}$

D $8\pi\ \text{ft}^3/\text{min}$

QUESTION 21

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A snowball melts so that $dV/dt = -4\pi$. If $V = (4/3)\pi r^3$, then dr/dt when $r = 2$ is

A -1

B -2

C $-1/4$

D $-1/2$

QUESTION 22

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A 13-ft ladder leans against a wall. The bottom slides away at 2 ft/s. When bottom is 5 ft from the wall, the top slides down at

A) 2 ft/s
B) 5/12 ft/s
C) 12/5 ft/s
D) 5/6 ft/s

QUESTION 23

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A balloon's radius increases at 0.5 cm/s. The rate of change of surface area $A = 4\pi r^2$ when $r = 10$ is

A) 20π
B) 40π
C) 80π
D) 10π

QUESTION 24

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Car A is 3 mi north of a junction, heading north at 40 mph. Car B is 4 mi east, heading east at 30 mph. How fast is the distance between them increasing?

A) 48 mph
B) 70 mph
C) 50 mph
D) 25 mph

QUESTION 25

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A cone has height always equal to 3 times the radius. If r increases at 2 in/s, dV/dt when r = 3 is

A) 108π

B) 54π

C) 18π

D) 36π

QUESTION 26

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The linearization L(x) of f at x = a is

A) $L(x) = f'(a) + f(a)(x - a)$

B) $L(x) = f(a)x$

C) $L(x) = f'(a)(x - a)$

D) $L(x) = f(a) + f'(a)(x - a)$

QUESTION 27

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Using the tangent line to $f(x) = \sqrt{x}$ at x = 4, $\sqrt{4.1}$ is approximately

A) 2.025

B) 2.1

C) 1.975

D) 2.05

QUESTION 28

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The linear approximation of e^x at $x = 0$ gives e^0.2 ≈

A 1.4

B 0.8

C 1.2

D 1.02

QUESTION 29

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

ln(1.05) is best approximated by the tangent line to ln x at $x = 1$ as

A 1.05

B 0.05

C 0.5

D 0.005

QUESTION 30

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f(3) = 5$ and $f'(3) = -2$, then $f(3.1) \approx$

A 3.1

B 4.8

C 4.9

D 5.2

QUESTION 31

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $f''(x) < 0$ near $x = a$, the tangent approximation $L(a + h)$ for small $h > 0$ is an

A exact value

B overestimate

C undefined

D underestimate

QUESTION 32

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f''(x) > 0$ near a, $L(a + h)$ is an

A equal to $f(a + h)$

B overestimate of $f(a + h)$

C equal to $f'(a)$

D underestimate of $f(a + h)$

QUESTION 33

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $y = x^3$, the differential dy equals

A $x^3\,dx$

B $3x^2\,dx$

C $3x\,dx$

D $3x^2$

QUESTION 34

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For $y = \sqrt{x}$, an estimate of the change in y when x increases from 9 to 9.3 is

A 0.3

B 0.025

C 0.05

D 0.15

QUESTION 35

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

L'Hospital's Rule may be applied directly to $\lim_{x \to a} [f(x)/g(x)]$ when the form is

A $0 \cdot \infty$ only

B any indeterminate form

C 1^{∞} only

D $0/0$ or $\pm\infty/\pm\infty$

QUESTION 36

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} \sin(5x)/x =$

A ∞

B 0

C 1

D 5

QUESTION 37

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (1 - \cos x)/x^2 =$

A) 0

B) 1

C) 1/2

D) ∞

QUESTION 38

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to \infty} (\ln x)/x =$

A) 1

B) e

C) ∞

D) 0

QUESTION 39

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to \infty} x / e^x =$

A) e

B) 1

C) ∞

D) 0

QUESTION 40

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0^+} x \ln x =$

A 1

B $-\infty$

C 0

D ∞

QUESTION 41

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} (e^{2x} - 1)/x =$

A 0

B 1

C 2

D e

QUESTION 42

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to \infty} (3x^2 + 1)/(5x^2 - 2)$ by applying L'Hospital's Rule twice equals

A ∞

B 3/5

C 3/2

D 0

QUESTION 43

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to 1} (x^3 - 1)/(x - 1) =$

A ∞

B 1

C 3

D 0

QUESTION 44

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} (\tan x)/x =$

A 1

B 0

C ∞

D π

QUESTION 45

HARD · MULTIPLE CHOICE

★★★ MANDATORY

Which of the following limits is NOT an indeterminate form 0/0 or ∞/∞ as written?

A $\lim_{x \to 0} (\sin x)/x$

B $\lim_{x \to \infty} x^2 / e^x$

C $\lim_{x \to 0} (\cos x)/x$

D $\lim_{x \to \infty} (\ln x)/\sqrt{x}$

QUESTION 46

EASY · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

In any related-rates problem, the first mathematical step is to

A substitute specific numbers immediately

B write an equation relating the variables, then differentiate with respect to time

C set the derivative equal to zero

D integrate both sides

QUESTION 47

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If the area of a square is increasing at 8 cm^2/s, then when side $s = 4$ cm, $ds/dt =$

A 8 cm/s

B 1 cm/s

C 1/2 cm/s

D 2 cm/s

QUESTION 48

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle moves with position $s(t) = \sin t$. Its maximum speed on $[0, 2\pi]$ is

A 0

B π

C 1

D 2

QUESTION 49

EASY · MULTIPLE CHOICE

★★ IMPORTANT

If f is differentiable and $f(2) = 3$, $f'(2) = 4$, the equation of the tangent line at $x = 2$ is

A $y - 3 = 2(x - 4)$

B $y = 4x$

C $y - 3 = 4(x - 2)$

D $y - 2 = 4(x - 3)$

QUESTION 50

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle moves along a line with $v(t) = 6t - t^2$. It is moving right when

A $0 < t < 6$

B $t > 6$

C $t = 3$ only

D $t < 0$

QUESTION 51

EASY · MULTIPLE CHOICE

★★ IMPORTANT

In a related-rates balloon problem, the physical meaning of dV/dt is

A the total volume

B the rate of change of the radius

C the rate of change of the volume

D the surface area

QUESTION 52

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Applying L'Hospital's Rule, $\lim_{x\to 0} (x - \sin x)/x^3 =$

A. 1/6

B. ∞

C. 1/2

D. 0

QUESTION 53

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A circular oil slick spreads so that $dr/dt = 0.1$ m/min. When $r = 50$ m, $dA/dt =$

A. 5π m²/min

B. 100π m²/min

C. 0.1π m²/min

D. 10π m²/min

QUESTION 54

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $y = f(x)$ and x is a differentiable function of t, then $dy/dt =$

A. $f'(x)$

B. $f'(x) \cdot (dx/dt)$

C. $(dx/dt) / f'(x)$

D. $f(x) \cdot (dx/dt)$

QUESTION 55

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A local linear approximation L(x) is most accurate when

A. $|x - a|$ is large

B. $|f''|$ is very large

C. f is non-differentiable at a

D. $|x - a|$ is small and $|f''|$ is small near a

QUESTION 56

HARD · MULTIPLE CHOICE

★★★ MANDATORY

A particle moves with $v(t) = t^3 - t$. It changes direction when

A. $t = 1$ only

B. $t = -1$, 0, and 1

C. $t = 0$ only

D. $t = \pm 1$ only

QUESTION 57

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $f(10) = 20$ and the differential $df = 0.4$ when $dx = 0.1$, then $f'(10) =$

A. 4

B. 40

C. 2

D. 0.4

QUESTION 58

HARD · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\infty} (\ln x)^2 / x =$

A. 0
B. 2
C. 1
D. ∞

QUESTION 59

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A trough is 10 ft long with triangular ends of height 3 ft and width 2 ft. If water rises at 0.2 ft/min when depth is 1.5 ft, dV/dt =

A. 4/3 ft^3/min
B. 2 ft^3/min
C. 1 ft^3/min
D. 0.5 ft^3/min

QUESTION 60

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The physical interpretation of the definite integral ∫_a^b v(t) dt is

A. acceleration change
B. total distance traveled
C. net change in position (displacement)
D. average speed

QUESTION 61

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A particle's speed is increasing if and only if

A. acceleration $a(t) > 0$

B. velocity $v(t) > 0$

C. velocity and acceleration have the same sign

D. $s''(t) = 0$

QUESTION 62

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} (\tan 3x) / (\sin 5x) =$

A. 5/3

B. 1

C. 3/5

D. 0

QUESTION 63

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

When using linearization, $f(a + \Delta x) \approx f(a) + f'(a)\Delta x$. This statement is equivalent to

A. the Mean Value Theorem

B. L'Hospital's Rule

C. the definition of the differential approximation

D. the Intermediate Value Theorem

QUESTION 64

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A spherical balloon is inflated so that $dV/dt = 8\pi$ in^3/s. When $r = 2$ in, $dr/dt =$

A. 1/4 in/s
B. 2 in/s
C. 1 in/s
D. 1/2 in/s

QUESTION 65

EASY · MULTIPLE CHOICE ★★★ MANDATORY

If $T(t)$ is temperature and $T'(t)$ is in °F per hour, then $T'(2) = -3$ means

A. At $t = 2$ hours the temperature is dropping at 3°F per hour
B. At $t = 2$ hours the temperature is −3°F
C. Temperature is increasing
D. The average temperature drop over 2 hours is 3°F

QUESTION 66

HARD · MULTIPLE CHOICE ★★★ MANDATORY

$\lim_{x \to \pi/2^-} (1 - \sin x)/(1 + \cos 2x) =$

A. ∞
B. 1/2
C. 1/4
D. 0

QUESTION 67

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A particle has velocity $v(t) = 4 - t$. From $t = 0$ to $t = 6$, the total distance traveled is

A 6

B 10

C 8

D 12

QUESTION 68

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A 6-ft-tall person walks away from a 15-ft streetlight at 4 ft/s. How fast is the tip of the shadow moving?

A 10 ft/s

B 20/3 ft/s

C 8/3 ft/s

D 4 ft/s

QUESTION 69

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A rocket launches vertically and is tracked by a camera 3000 ft from the launch pad. When the rocket is 4000 ft high, rising at 800 ft/s, how fast is the camera's angle of elevation changing?

A 0.128 rad/s

B 0.096 rad/s

C 0.048 rad/s

D 0.240 rad/s

QUESTION 70

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Using the tangent line approximation to $f(x) = \cos x$ at $x = 0$, the approximation for $\cos(0.1)$ is

A 0.995, and it is exact

B 1, and it is an underestimate

C 0.9, and it is an underestimate

D 1, and it is an overestimate

QUESTION 71

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The length of a rectangle increases at 3 cm/s and its width decreases at 2 cm/s. When length = 8 cm and width = 5 cm, the rate of change of area is

A $1\ \text{cm}^2/\text{s}$

B $-1\ \text{cm}^2/\text{s}$

C $14\ \text{cm}^2/\text{s}$

D $-26\ \text{cm}^2/\text{s}$

QUESTION 72

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A particle moves with position $s(t) = 2t^3 - 9t^2 + 12t$. On which interval is the particle's speed decreasing?

A $(0, 1)$ and $(1.5, 2)$

B $(1, 2)$

C $(0, 1.5)$

D $(1, 1.5)$ and $(2, \infty)$

QUESTION 73

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\lim_{x\to\infty} x^3 / e^{2x} =$

A ∞

B 0

C 6

D 1/8

QUESTION 74

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\lim_{x\to 0} (\sqrt{(4+3x)} - 2) / x =$

A 3/2

B 0

C 3/4

D 1/4

QUESTION 75

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The side of a square is measured as 10 cm with maximum error ±0.1 cm. The maximum relative error in the calculated area is

A 0.02 (2%)

B 0.10 (10%)

C 0.04 (4%)

D 0.01 (1%)

QUESTION 76

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Total revenue is $R(x) = 50x - 0.5x^2$ and total cost is $C(x) = 10x + 100$. Profit is maximized at production level $x =$

A 50

B 40

C 30

D 100

QUESTION 77

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

A particle moves with $v(t) = \pi \cos(\pi t)$ on $[0, 2]$. The total distance traveled is

A 0

B 2

C 4

D 2π

QUESTION 78

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A cylindrical tank of constant radius $R = 3$ m is draining so height drops at 0.5 m/min. How fast is volume decreasing?

A 0.5π m³/min

B 1.5π m³/min

C 9π m³/min

D 4.5π m³/min

QUESTION 79

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} (x - \tan x) / x^3 =$

A. $1/3$

B. 0

C. -1

D. $-1/3$

QUESTION 80

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $P(t)$ be price index. If $P'(t) > 0$ and $P''(t) < 0$, which describes prices?

A. Prices are increasing at an accelerating rate

B. Prices are increasing, but the rate of inflation is slowing down

C. Prices are constant

D. Prices are falling

QUESTION 81

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Using the tangent line to $f(x) = \sqrt[3]{x}$ at $x = 27$, $\sqrt[3]{28}$ is approximately

A. 3.050

B. $3 + 1/81 \approx 3.012$

C. $3 + 1/9 \approx 3.111$

D. $3 + 1/27 \approx 3.037$

QUESTION 82

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Water pours into an inverted conical tank (height 10 ft, radius 5 ft) at 6 ft^3/min. When water depth is 6 ft, dh/dt =

A $6/\pi$ ft/min

B $1/(6\pi)$ ft/min

C $2/(3\pi)$ ft/min

D $3/(2\pi)$ ft/min

QUESTION 83

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle moves with position $s(t) = t^4 - 6t^2 + 5$. At what time $t > 0$ is acceleration zero?

A $t = 1$

B $t = 2$

C $t = 1/2$

D $t = \sqrt{3}$

QUESTION 84

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} (1 - \cos(4x)) / (x \sin(2x)) =$

A 0

B 2

C 4

D 8

QUESTION 85

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Coffee cools according to T(t) = 20 + 70e^(−0.05t) °C. At t = 0, the rate of cooling is

A 0.05 °C/min

B 20 °C/min

C 70 °C/min

D 3.5 °C/min

QUESTION 86

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Ship A sails south at 15 knots and Ship B sails west at 20 knots from the same port starting at noon. At 2:00 PM, the distance between them increases at

A 35 knots

B 12.5 knots

C 50 knots

D 25 knots

QUESTION 87

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0^+} (\sin x)(\ln x) =$

A 1

B −1

C $-\infty$

D 0

QUESTION 88

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Using the linearization of $f(x) = \sin x$ at $a = 0$, $\sin(0.05) \approx 0.05$. This approximation is an

A overestimate

B undefined value

C exact value

D underestimate

QUESTION 89

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A particle has velocity $v(t) = 2t^3 - 3t$. Its average acceleration on $[1, 3]$ is

A 30

B 23

C 15

D 46

QUESTION 90

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The side of an equilateral triangle increases at $\sqrt{3}$ cm/s. When each side is 4 cm, the area increases at

A $12\ \text{cm}^2/\text{s}$

B $4\sqrt{3}\ \text{cm}^2/\text{s}$

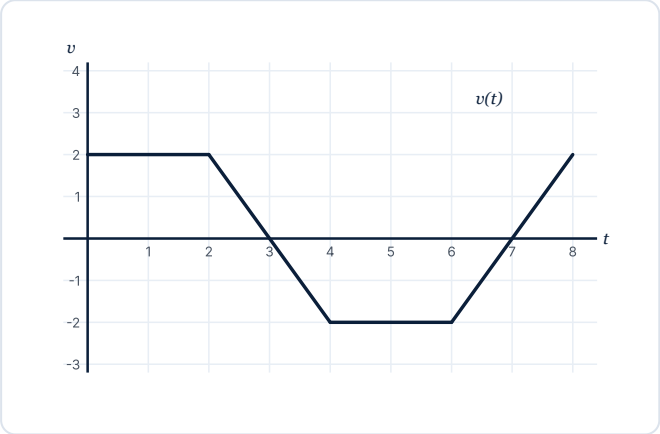
C $2\sqrt{3}\ \text{cm}^2/\text{s}$

D $6\ \text{cm}^2/\text{s}$

Question 91: Particle Motion from a Velocity Graph

★★★★★ VERY IMPORTANT 9 Points

The graph of the velocity v(t), in meters per second, of a particle moving along a horizontal line is shown below for $0 \le t \le 8$ seconds. The particle is at position $s(0) = 3$ meters at time $t = 0$.



Velocity v(t) of a particle, in m/s, for $0 \le t \le 8$ seconds

(a) [2 points] Find the acceleration of the particle at t = 2.5 seconds and at t = 5 seconds. Include appropriate units.

(b) [2 points] For what values of t in the open interval (0, 8) is the particle moving to the left? Justify your answer.

(c) [3 points] Find the total distance traveled by the particle over the time interval [0, 8].

(d) [2 points] Find the position of the particle at time t = 8 seconds.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 92: Related Rates: Sliding Ladder

★★★★★ VERY IMPORTANT 9 Points

A 13-foot ladder leans against a vertical wall. The bottom of the ladder slides away from the wall along the horizontal floor at a constant rate of 2 feet per second. Let x represent the distance from the wall to the base of the ladder, and let y represent the height of the top of the ladder above the floor.

(a) [2 points] Write an equation relating the variables x and y.

(b) [4 points] Find dy/dt at the instant when the base of the ladder is 5 feet from the wall.

(c) [3 points] At that instant, state whether the top of the ladder is moving up or down, and determine the rate at which the area of the triangle formed by the ladder, wall, and floor is changing.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 93: Linear Approximation and Concavity

★★★★★ VERY IMPORTANT 9 Points

Let f be the function defined by $f(x) = \sqrt{x}$.

(a) [3 points] Write an equation for the linearization L(x) of f at a = 9.

(b) [3 points] Use L(x) to approximate the value of $\sqrt{9.3}$.

(c) [3 points] Determine whether the approximation found in part (b) is an overestimate or an underestimate of the true value of $\sqrt{9.3}$. Justify your answer using the second derivative f''.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 94: Tabular Analysis: Interpretation and Tangent Line

★★★★★ VERY IMPORTANT 9 Points

A differentiable function f has values and derivatives given in the table below for selected values of x.

x	1	2	3	4
f(x)	5	8	10	11
f′(x)	4	3	2	0.5

(a) [2 points] If f(x) represents the temperature of an oven in degrees Celsius at time x hours, interpret the meaning of f′(2) = 3 in context, including units of measurement.

(b) [3 points] Write the equation of the tangent line to the graph of f at x = 2, and use it to approximate f(2.2).

(c) [2 points] Given that f″(x) < 0 for all x on the closed interval [1, 4], state whether your approximation in part (b) is an overestimate or an underestimate. Explain your reasoning.

(d) [2 points] Find the average rate of change of f over the interval [1, 4]. Include appropriate units.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 95: Related Rates: Expanding Spherical Balloon

★★★★★ VERY IMPORTANT 9 Points

A spherical balloon is being inflated with helium so that its volume increases at a constant rate of 8π cubic inches per second. The volume of a sphere of radius r is given by $V = (4/3)\pi r^3$.

(a) [3 points] Find an expression for dr/dt in terms of the radius r.

(b) [3 points] Find the rate of change of the radius dr/dt at the instant when r = 2 inches. Include units.

(c) [3 points] Find the rate of change of the surface area dA/dt at the instant when r = 2 inches, given $A = 4\pi r^2$.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 96: Straight-Line Motion from an Algebraic Formula

★★★★★ VERY IMPORTANT 9 Points

A particle moves along the x-axis so that its position at any time $t \geq 0$ is given by $s(t) = t^3 - 6t^2 + 9t$, where s is measured in meters and t is in seconds.

(a) [2 points] Find the velocity v(t) and acceleration a(t) of the particle as functions of t.

(b) [2 points] Determine all values of t for which the particle is at rest.

(c) [3 points] Find the total distance traveled by the particle over the time interval [0, 4].

(d) [2 points] At t = 2.5 seconds, is the speed of the particle increasing or decreasing? Justify your answer.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 97: Indeterminate Forms and L'Hospital's Rule

★★★★★ VERY IMPORTANT | 9 Points

Evaluate each of the following limits using L'Hospital's Rule. For each part, explicitly state the indeterminate form before applying the rule, and show all analytical steps.

(a) [3 points] $\lim_{x \to 0} \frac{e^{3x} - 1}{\sin(2x)}$

(b) [3 points] $\lim_{x \to \infty} \frac{\ln x}{\sqrt{x}}$

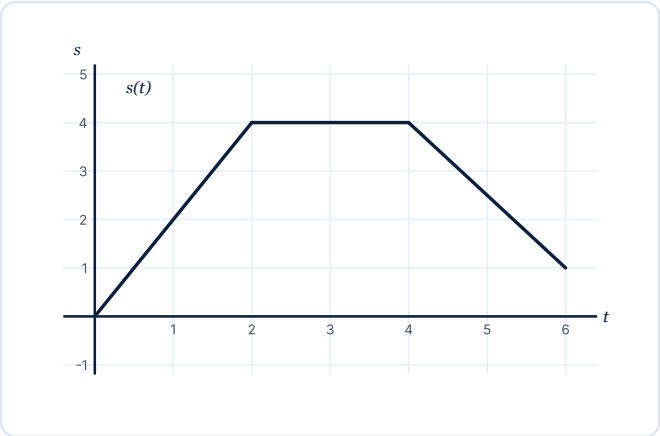
(c) [3 points] $\lim_{x \to 0^+} x \ln x$

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 98: Analysis of Motion from a Position Graph

★★★★★ VERY IMPORTANT 9 Points

The graph of the position function s(t), in meters, of a particle moving along a coordinate line is shown below for $0 \le t \le 6$ seconds.



Position s(t) of a particle, in meters, for $0 \le t \le 6$

(a) [2 points] Find the average velocity on [0, 2] and the instantaneous velocity at t = 5 seconds.

(b) [2 points] On which interval is the particle at rest? Justify your answer.

(c) [2 points] On which interval is the particle moving in the negative direction (to the left)? Justify your answer.

(d) [3 points] Find the total distance traveled by the particle over [0, 6] and the net displacement s(6) − s(0).

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 99: Applied Rate of Change from Tabular Data

★★★ MANDATORY

9 Points

Water is leaking out of a storage container. The volume of water remaining in the container, V(t), in gallons, is recorded at selected times t, in minutes, as shown in the table below.

t (min)	0	5	10	15
V(t) (gal)	120	95	78	66

(a) [3 points] Estimate the value of $V'(10)$ using a symmetric difference quotient from the table. Include appropriate units.

(b) [2 points] Interpret the meaning of your answer to part (a) in the context of the problem.

(c) [2 points] Find the average rate of change of the volume over the entire 15-minute time period. Include units.

(d) [2 points] Is the water leaking faster during the first 5 minutes or during the last 5 minutes? Justify using values from the table.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

Question 100: Related Rates: Inverted Conical Reservoir

★★★★★ VERY IMPORTANT 9 Points

An inverted conical reservoir (vertex pointing downwards) has a height of 12 feet and a radius of 3 feet at the top. Water flows into the reservoir at a constant rate of 10 cubic feet per minute. Let h be the depth of the water in feet, and let r be the radius of the water's surface in feet.

(a) [3 points] Use similar triangles to express r in terms of h, and express the volume of water V as a function of h alone.

(b) [4 points] Find the rate at which the water level is rising, dh/dt, at the instant when the water depth is h = 4 feet.

(c) [2 points] Is the water level rising faster when h = 4 feet or when h = 8 feet? Justify your answer analytically.

Reminder: Show related-rates equations, units of measurement, and sign analysis for particle motion.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point rubrics.
- **Appendix:** Essential Unit 4 formulas, rules, and theorems.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
B ★★★★★	D ★★	B ★★★★★	C ★★★★★	A ★★	C ★★★	A ★★★	C ★★★★★	B ★★★★★	C ★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
C ★★★★★	B ★★★	B ★★★★★	B ★★★	B ★★★★★	D ★★★★★	C ★★★	B ★★★★★	D ★★★	B ★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
C ★★★★★	D ★★★★★	B ★★★	A ★★★★★	B ★★★★★	D ★★★★★	A ★★★★★	C ★★★	B ★★★	B ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
B ★★★★★	D ★★★	B ★★★	C ★★★	D ★★★★★	D ★★★	C ★★★★★	D ★★★★★	D ★★★★★	C ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
C ★★★	B ★★	C ★★	A ★★★	C ★★★	B ★★★★★	B ★★★	C ★★★	C ★★	A ★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★	A ★★★★★	D ★★★	B ★★★★★	D ★★★	B ★★★	A ★★★	A ★★★	B ★★★★★	C ★★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
C ★★★★★	C ★★★	C ★★	D ★★★	A ★★★	C ★★★	B ★★★★★	B ★★★★★	B ★★★★★	D ★★★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
B ★★★	A ★★★★★	B ★★★	C ★★★	A ★★★	B ★★★	C ★★★★★	D ★★★	D ★★★★★	B ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
D ★★★	C ★★★★★	A ★★★	C ★★★★★	D ★★★	D ★★★★★	D ★★★★★	A ★★★★★	B ★★★	D ★★★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01

★★★★★ VERY IMPORTANT

Correct Answer: (B) In 2014 the population is decreasing at 250 people per year

STEP-BY-STEP SOLUTION

P′(4) is the instantaneous rate of change of population at t = 4 (year 2014). The units are people per year. The negative sign means the population is decreasing.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** P′ is a rate, not a population value.

✗ **(C)** This is an average rate over 4 years, not an instantaneous rate.

✗ **(D)** This confuses P(4) with P′(4).

Question 02

★★ IMPORTANT

Correct Answer: (D) liters per minute

STEP-BY-STEP SOLUTION

The derivative of a quantity with respect to time has units of that quantity divided by time: liters/minute.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Those are the units of V, not V′.

✗ **(B)** Those are the units of t.

✗ **(C)** Those would be the units of V″.

Question 03

★★★★★ VERY IMPORTANT

Correct Answer: (B) The 51st item costs about $3.20 to produce

STEP-BY-STEP SOLUTION

C′(50) is the marginal cost at 50 items: the approximate extra cost of producing one more item.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Marginal cost is not profit.

✗ **(C)** That would be C(50).

✗ **(D)** Average cost is C(x)/x.

Question 04

★★★★★ VERY IMPORTANT

Correct Answer: (C) T is increasing, but the rate of increase is slowing

STEP-BY-STEP SOLUTION

$T'(10) > 0$ so temperature is increasing. $T''(10) < 0$ so T' is decreasing: the increase is slowing (concave down).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** T' is positive, so T is not decreasing.

✗ **(B)** $T' \neq 0$.

✗ **(D)** $T'' < 0$ means the rate is not speeding up.

Question 05

★★ IMPORTANT

Correct Answer: (A) m/s^2

STEP-BY-STEP SOLUTION

s'' is acceleration: meters per second squared.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Those are velocity units.

✗ **(C)** Those are position units.

✗ **(D)** Inverted units.

Question 06

★★★ MANDATORY

Correct Answer: (C) t = 1 and t = 3

STEP-BY-STEP SOLUTION

At rest means $v(t) = 0$: $3(t^2 - 4t + 3) = 3(t - 1)(t - 3) = 0$, so $t = 1$ and $t = 3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Misses $t = 3$.

✗ **(B)** Misses $t = 1$.

✗ **(D)** $v(0) = 9 \neq 0$.

Question 07

★★★ MANDATORY

Correct Answer: (A) −6

STEP-BY-STEP SOLUTION

$v(t) = 3t^2 - 12t + 9$, $a(t) = 6t - 12$. At $t = 1$, $a(1) = 6(1) - 12 = -6$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Confusion with velocity coefficients.

✗ **(C)** This is $a(2)$.

✗ **(D)** Sign error.

Question 08

★★★★★ VERY IMPORTANT

Correct Answer: (C) 5

STEP-BY-STEP SOLUTION

Speed is $|v|$. $v(3) = 9 - 4 = 5$, so speed $= |5| = 5$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $v(3)$ is not 1.

✗ **(B)** This is t^2.

✗ **(D)** Speed is never negative.

Question 09

★★★★★ VERY IMPORTANT

Correct Answer: (B) slowing down and moving left

STEP-BY-STEP SOLUTION

$v < 0$ means moving left (negative direction). Opposite signs of v and a mean speed is decreasing (slowing down).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** v is negative, so not moving right.

✗ **(C)** $v \neq 0$, so not at rest.

✗ **(D)** Opposite signs mean slowing down, not speeding up.

Question 10

★★★ MANDATORY

Correct Answer: (C) speeding up in the positive direction

STEP-BY-STEP SOLUTION

Same signs of v and a imply speed is increasing. v > 0 means movement in the positive direction.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** v > 0 is the positive direction.

✗ **(B)** Same signs mean speeding up, not slowing down.

✗ **(D)** No sign change of v is indicated.

Question 11

★★★★★ VERY IMPORTANT

Correct Answer: (C) t = 2

STEP-BY-STEP SOLUTION

v(t) = 2t − 4 = 0 at t = 2. v changes from negative to positive at t = 2, so the particle changes direction there.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** s(3)=0 is not a direction change.

✗ **(B)** s(1)=0 is a position, not a velocity zero.

✗ **(D)** Those are roots of s(t), not of v(t).

Question 12

★★★ MANDATORY

Correct Answer: (B) 5

STEP-BY-STEP SOLUTION

[s(4) − s(1)] / (4 − 1) = (16 − 1) / 3 = 15 / 3 = 5.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses s(4) − s(1) without dividing by Δt.

✗ **(C)** Uses only the time span 3.

✗ **(D)** Arithmetic error.

Question 13

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\lim_{h\to 0} [s(a+h) - s(a)] / h$

STEP-BY-STEP SOLUTION

Instantaneous velocity is $s'(a)$, defined as the limit of the average velocity difference quotient as h approaches 0.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is position.

✗ **(C)** This is average velocity over an interval of length h.

✗ **(D)** This is acceleration.

Question 14

★★★ MANDATORY

Correct Answer: (B) 16

STEP-BY-STEP SOLUTION

$v(t) = \int 6t\, dt = 3t^2 + C$. $v(0) = 4 \Rightarrow C = 4$. Therefore $v(2) = 3(4) + 4 = 16$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses only $v(0)$ without integrating.

✗ **(C)** Forgets the initial velocity $v(0) = 4$.

✗ **(D)** Computes $6t^2$ at $t = 2$.

Question 15

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\int_0^3 v(t)\, dt$

STEP-BY-STEP SOLUTION

Net displacement is the definite integral of velocity: $s(3) - s(0) = \int_0^3 v(t)\, dt$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is total distance traveled.

✗ **(C)** This is net change in velocity.

✗ **(D)** This is the initial position.

Question 16

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\int_a^b |v(t)|\,dt$

STEP-BY-STEP SOLUTION

Total distance accounts for every unit of motion forward and backward, so one integrates speed $|v(t)|$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is displacement magnitude, which cancels back-and-forth travel.

✗ **(B)** This is net displacement.

✗ **(C)** This ignores internal turning points.

Question 17

★★★ MANDATORY

Correct Answer: (C) 6 gallons per minute

STEP-BY-STEP SOLUTION

$V'(3) = -6$ gal/min. The word "leaking" implies the rate of volume loss, which is 6 gallons per minute.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses t = 3 directly without multiplying by 2.

✗ **(B)** This is $|V'|$ at t = 1.

✗ **(D)** Rates must include time units (gal/min).

Question 18

★★★★★ VERY IMPORTANT

Correct Answer: (B) 20π cm²/s

STEP-BY-STEP SOLUTION

$A = \pi r^2 \Rightarrow dA/dt = 2\pi r\,(dr/dt) = 2\pi(5)(2) = 20\pi$ cm²/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is πr^2 without differentiating.

✗ **(C)** Forgets the radius r.

✗ **(D)** Omits a factor of 2.

Question 19

★★★ MANDATORY

Correct Answer: (D) 36 cm³/s

STEP-BY-STEP SOLUTION

$V = x^3 \Rightarrow dV/dt = 3x^2\,(dx/dt) = 3(2^2)(3) = 3(4)(3) = 36\ \text{cm}^3/\text{s}$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is x^3.

✗ **(B)** Uses $3x^2$ without multiplying by $dx/dt = 3$.

✗ **(C)** Uses $3x\ dx/dt$.

Question 20

★★★ MANDATORY

Correct Answer: (B) 32π ft³/min

STEP-BY-STEP SOLUTION

$V = \pi r^2 h = 16\pi\ h$. Differentiating gives $dV/dt = 16\pi\,(dh/dt) = 16\pi(2) = 32\pi\ \text{ft}^3/\text{min}$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits the radius squared.

✗ **(C)** Forgets the rate $dh/dt = 2$.

✗ **(D)** Uses $\pi r\ dh/dt$.

Question 21

★★★★★ VERY IMPORTANT

Correct Answer: (C) −1/4

STEP-BY-STEP SOLUTION

$dV/dt = 4\pi r^2\,(dr/dt)$. $-4\pi = 4\pi(2^2)(dr/dt) = 16\pi\,(dr/dt) \Rightarrow dr/dt = -4\pi\ /\ (16\pi) = -1/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Cancels 4π incorrectly.

✗ **(B)** Uses r instead of r^2.

✗ **(D)** Forgets that $r^2 = 4$.

Question 22

★★★★★ VERY IMPORTANT

Correct Answer: (D) 5/6 ft/s

STEP-BY-STEP SOLUTION

$x^2 + y^2 = 169$. When $x = 5$, $y = 12$. Differentiating: $2x(dx/dt) + 2y(dy/dt) = 0 \Rightarrow dy/dt = -(x/y)(dx/dt) = -(5/12)(2) = -5/6$. The top slides down at 5/6 ft/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is dx/dt.

✗ **(B)** Forgets to multiply by dx/dt = 2.

✗ **(C)** This is (y/x)(dx/dt).

Question 23

★★★ MANDATORY

Correct Answer: (B) 40π

STEP-BY-STEP SOLUTION

$dA/dt = 8\pi r\,(dr/dt) = 8\pi(10)(0.5) = 40\pi$ cm²/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses $4\pi r\,(dr/dt)$.

✗ **(C)** Omits dr/dt = 0.5.

✗ **(D)** Uses r (dr/dt).

Question 24

★★★★★ VERY IMPORTANT

Correct Answer: (A) 48 mph

STEP-BY-STEP SOLUTION

$s^2 = x^2 + y^2$. At this instant $x = 4$, $y = 3$, so $s = 5$. $2s(ds/dt) = 2x(dx/dt) + 2y(dy/dt) \Rightarrow 5(ds/dt) = 4(30) + 3(40) = 120 + 120 = 240 \Rightarrow ds/dt = 48$ mph.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Simply adds the two speeds 40 + 30.

✗ **(C)** Assumes the rate equals the hypotenuse value.

✗ **(D)** Swaps coordinates and rates.

Question 25

★★★★★ VERY IMPORTANT

Correct Answer: (B) 54π

STEP-BY-STEP SOLUTION

h = 3r, so V = (1/3)πr²(3r) = πr³. dV/dt = 3πr² (dr/dt) = 3π(9)(2) = 54π in³/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Multiplies by an extra factor of 2.

✗ **(C)** Omits dr/dt = 2.

✗ **(D)** Uses r = 3 only once.

Question 26

★★★★★ VERY IMPORTANT

Correct Answer: (D) L(x) = f(a) + f′(a)(x – a)

STEP-BY-STEP SOLUTION

By definition, L(x) is the tangent line equation at (a, f(a)): L(x) = f(a) + f′(a)(x – a).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Swaps f(a) and f′(a).

✗ **(B)** Linear through origin, incorrect.

✗ **(C)** Omits the base value f(a).

Question 27

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2.025

STEP-BY-STEP SOLUTION

f(4) = 2, f′(x) = 1/(2√x) ⇒ f′(4) = 1/4. L(4.1) = 2 + (1/4)(0.1) = 2 + 0.025 = 2.025.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Adds Δx = 0.1 directly to 2.

✗ **(C)** Subtracts the linear increment instead of adding.

✗ **(D)** Uses f′(4) = 1/2.

Question 28

★★★ MANDATORY

Correct Answer: (C) 1.2

STEP-BY-STEP SOLUTION

$f(0) = 1$, $f'(x) = e^x \Rightarrow f'(0) = 1$. $L(0.2) = 1 + 1(0.2) = 1.2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Doubles the increment.

✗ **(B)** Subtracts 0.2 from 1.

✗ **(D)** Misplaces the decimal place as 0.02.

Question 29

★★★ MANDATORY

Correct Answer: (B) 0.05

STEP-BY-STEP SOLUTION

$f(1) = 0$, $f'(x) = 1/x \Rightarrow f'(1) = 1$. $L(1.05) = 0 + 1(0.05) = 0.05$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the input value x.

✗ **(C)** Off by a factor of 10.

✗ **(D)** Off by a factor of 1/10.

Question 30

★★★ MANDATORY

Correct Answer: (B) 4.8

STEP-BY-STEP SOLUTION

$L(3.1) = f(3) + f'(3)(0.1) = 5 + (-2)(0.1) = 5 - 0.2 = 4.8$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Reports the input x-value.

✗ **(C)** Calculates Δx as 0.05.

✗ **(D)** Adds 0.2 instead of subtracting.

Question 31

★★★★★ VERY IMPORTANT

Correct Answer: (B) overestimate

STEP-BY-STEP SOLUTION

$f'' < 0$ means f is concave down, so the curve lies entirely below its tangent line. Therefore, L(x) overestimates f(x).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** A tangent line is never identically equal to a curved graph.

✗ **(C)** Linear approximations are well-defined for differentiable functions.

✗ **(D)** Concave up yields underestimates.

Question 32

★★★ MANDATORY

Correct Answer: (D) underestimate of f(a + h)

STEP-BY-STEP SOLUTION

$f'' > 0$ means f is concave up, so the curve lies above its tangent line. Thus, L(x) underestimates f(x).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Only equal at the point of tangency $h = 0$.

✗ **(B)** Concave down gives overestimates.

✗ **(C)** L is an approximation to function value f, not f'.

Question 33

★★★ MANDATORY

Correct Answer: (B) $3x^2\,dx$

STEP-BY-STEP SOLUTION

By definition of the differential, $dy = f'(x)\,dx = 3x^2\,dx$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the original function instead of the derivative.

✗ **(C)** Incorrect power of x.

✗ **(D)** Omits the differential dx.

Question 34

★★★ MANDATORY

Correct Answer: (C) 0.05

STEP-BY-STEP SOLUTION

dy = [1 / (2√x)] dx. At x = 9, dx = 0.3: dy = [1 / (2·3)] (0.3) = 0.3 / 6 = 0.05.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Simply takes dx = 0.3.

✗ **(B)** Divides by 12 instead of 6.

✗ **(D)** Uses 1 / √x without the factor 1/2.

Question 35

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0/0 or ±∞/±∞

STEP-BY-STEP SOLUTION

L'Hospital's Rule requires an indeterminate quotient 0/0 or ±∞/±∞.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** 0 · ∞ must first be converted into a quotient.

✗ **(B)** Other indeterminate forms must be rewritten before applying L'H.

✗ **(C)** 1^∞ requires logarithmic transformation first.

Question 36

★★★ MANDATORY

Correct Answer: (D) 5

STEP-BY-STEP SOLUTION

Form 0/0. Applying L'Hospital: $\lim$ (5 cos 5x)/1 = 5(1) = 5.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The limit is finite and well-defined.

✗ **(B)** Takes sin 0 = 0 in numerator.

✗ **(C)** Confuses with standard sin x / x.

Question 37

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1/2

STEP-BY-STEP SOLUTION

Form 0/0. L'H once: sin x / (2x) (still 0/0). L'H second time: cos x / 2 → 1/2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Stops after inspecting numerator only.

✗ **(B)** Omits the factor of 2 in the denominator.

✗ **(D)** Does not diverge to infinity.

Question 38

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

Form ∞/∞. Applying L'Hospital: $\lim (1/x)/1 = \lim 1/x = 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Assumes ln x and x grow at the same rate.

✗ **(B)** Irrelevant constant.

✗ **(C)** Polynomial growth strictly dominates logarithmic growth.

Question 39

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

Form ∞/∞. Applying L'Hospital: $\lim 1 / e^x = 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Irrelevant constant.

✗ **(B)** Equal rate of growth is false.

✗ **(C)** Exponential strictly dominates polynomial.

Question 40

★★★★★ VERY IMPORTANT

Correct Answer: (C) 0

STEP-BY-STEP SOLUTION

Form $0 \cdot (-\infty)$. Rewrite as $(\ln x) / (1/x)$, form $-\infty/\infty$. L'H: $(1/x) / (-1/x^2) = -x \to 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Arithmetic confusion.

✗ **(B)** ln x approaches $-\infty$, but the product is damped to 0.

✗ **(D)** Sign error in growth direction.

Question 41

★★★ MANDATORY

Correct Answer: (C) 2

STEP-BY-STEP SOLUTION

Form 0/0. L'Hospital's Rule gives ***lim*** [2e^(2x)] / 1 = 2(1) = 2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes numerator directly.

✗ **(B)** Omits the chain rule factor of 2.

✗ **(D)** Confuses e^0 with e.

Question 42

★★ IMPORTANT

Correct Answer: (B) 3/5

STEP-BY-STEP SOLUTION

First derivative: 6x / (10x). Second derivative: 6 / 10 = 3/5.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Does not diverge.

✗ **(C)** Misreads denominator coefficient.

✗ **(D)** Degrees are equal, limit is nonzero.

Question 43

★★ IMPORTANT

Correct Answer: (C) 3

STEP-BY-STEP SOLUTION

Form 0/0. L'Hospital's Rule gives $\lim 3x^2 / 1 = 3(1)^2 = 3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Indeterminate form is removable.

✗ **(B)** Uses the x-value.

✗ **(D)** Evaluates numerator as 0.

Question 44

★★★ MANDATORY

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

Form 0/0. L'Hospital gives $\lim \sec^2 x / 1 = \sec^2(0) = 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** tan 0 = 0 in numerator.

✗ **(C)** The limit is finite.

✗ **(D)** Trigonometric confusion.

Question 45

★★★ MANDATORY

Correct Answer: (C) $\lim_{x\to 0} (\cos x)/x$

STEP-BY-STEP SOLUTION

As $x \to 0$, $\cos(0)/0 = 1/0$, which is an undefined/infinite form, not 0/0 or ∞/∞. L'Hospital does not apply.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is 0/0.

✗ **(B)** This is ∞/∞.

✗ **(D)** This is ∞/∞.

Question 46

★★★★★ VERY IMPORTANT

Correct Answer: (B) write an equation relating the variables, then differentiate with respect to time

STEP-BY-STEP SOLUTION

One must relate the quantities geometrically or algebraically, differentiate implicitly with respect to t, and only then plug in instantaneous numbers.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Plugging in too early turns variables into constants with derivative 0.
✗ **(C)** Setting derivative to zero is for optimization.
✗ **(D)** Related rates requires differentiation, not integration.

Question 47

★★★ MANDATORY

Correct Answer: (B) 1 cm/s

STEP-BY-STEP SOLUTION

$A = s^2 \Rightarrow dA/dt = 2s\,(ds/dt)$. $8 = 2(4)(ds/dt) = 8\,(ds/dt) \Rightarrow ds/dt = 1$ cm/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sets ds/dt equal to dA/dt.
✗ **(C)** Inverts the fraction.
✗ **(D)** Forgets the factor of 2.

Question 48

★★★ MANDATORY

Correct Answer: (C) 1

STEP-BY-STEP SOLUTION

$v(t) = s'(t) = \cos t$. Speed is $|v(t)| = |\cos t|$, which has a maximum value of 1.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the minimum speed.
✗ **(B)** Trigonometric argument confusion.
✗ **(D)** This is the amplitude range of position.

Question 49

★★ IMPORTANT

Correct Answer: (C) $y - 3 = 4(x - 2)$

STEP-BY-STEP SOLUTION

Using point-slope form: $y - y_1 = m(x - x_1)$ with point (2, 3) and slope $m = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Swaps slope and x-coordinate.

✗ **(B)** Incorrect y-intercept.

✗ **(D)** Swaps the coordinates x and y.

Question 50

★★★ MANDATORY

Correct Answer: (A) $0 < t < 6$

STEP-BY-STEP SOLUTION

Moving right means $v(t) > 0$. $v(t) = t(6 - t) > 0$ on the interval (0, 6).

WHY THE OTHER OPTIONS FAIL

✗ **(B)** v is negative for $t > 6$.

✗ **(C)** $t = 3$ is where velocity is maximized, but motion is right throughout (0, 6).

✗ **(D)** v is negative for $t < 0$.

Question 51

★★ IMPORTANT

Correct Answer: (C) the rate of change of the volume

STEP-BY-STEP SOLUTION

V is volume and t is time, so dV/dt represents the instantaneous rate of change of volume over time.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** That is V.

✗ **(B)** That is dr/dt.

✗ **(D)** That is A.

Question 52

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1/6

STEP-BY-STEP SOLUTION

Form 0/0. L'H 1: $(1 - \cos x)/(3x^2)$. L'H 2: $\sin x / (6x)$. L'H 3: $\cos x / 6 \to 1/6$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The limit is finite.

✗ **(C)** Miscalculates the third derivative denominator.

✗ **(D)** Stops prematurely at 0.

Question 53

★★★ MANDATORY

Correct Answer: (D) 10π m²/min

STEP-BY-STEP SOLUTION

$A = \pi r^2 \Rightarrow dA/dt = 2\pi r\,(dr/dt) = 2\pi(50)(0.1) = 10\pi$ m²/min.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits the factor of 2.

✗ **(B)** Uses r^2 instead of 2r.

✗ **(C)** Uses dr/dt without radius scaling.

Question 54

★★★★★ VERY IMPORTANT

Correct Answer: (B) $f'(x) \cdot (dx/dt)$

STEP-BY-STEP SOLUTION

By the Chain Rule with respect to time: $dy/dt = (dy/dx) \cdot (dx/dt) = f'(x) \cdot (dx/dt)$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits the chain rule factor dx/dt.

✗ **(C)** Inverts the relationship.

✗ **(D)** Uses f instead of derivative f'.

Question 55

★★★ MANDATORY

Correct Answer: (D) $|x - a|$ is small and $|f''|$ is small near a

STEP-BY-STEP SOLUTION

The error term in linearization depends on the distance $|x - a|$ and the magnitude of the second derivative $|f''|$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Large distances increase approximation error.

✗ **(B)** High curvature increases approximation error.

✗ **(C)** f must be differentiable at a for L(x) to exist.

Question 56

★★★ MANDATORY

Correct Answer: (B) t = −1, 0, and 1

STEP-BY-STEP SOLUTION

$v(t) = t(t - 1)(t + 1)$. Testing intervals around −1, 0, 1 reveals that v changes sign across all three roots.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Misses t = 0 and t = −1.

✗ **(C)** Misses t = ±1.

✗ **(D)** Misses t = 0 where v also changes sign.

Question 57

★★★ MANDATORY

Correct Answer: (A) 4

STEP-BY-STEP SOLUTION

$df = f'(10)\,dx \Rightarrow 0.4 = f'(10)(0.1) \Rightarrow f'(10) = 0.4 / 0.1 = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Multiplies by 10 instead of dividing by 0.1.

✗ **(C)** Arithmetic slip.

✗ **(D)** Confuses df with f′(10).

Question 58

★★★ MANDATORY

Correct Answer: (A) 0

STEP-BY-STEP SOLUTION

Form ∞/∞. L'H 1: $[2(\ln x)(1/x)] / 1 = (2 \ln x)/x$. L'H 2: $(2/x)/1 = 2/x \to 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Arbitrary value.

✗ **(C)** Limits of logarithm/polynomial ratios do not yield 1.

✗ **(D)** Powers of ln x still grow slower than any positive power of x.

Question 59

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2 ft^3/min

STEP-BY-STEP SOLUTION

$w/h = 2/3 \Rightarrow w = (2/3)h$. Area $= (1/2)wh = (1/3)h^2$. Volume $= 10(1/3)h^2 = (10/3)h^2$. $dV/dt = (20/3)h\,(dh/dt) = (20/3)(1.5)(0.2) = 2\ ft^3/min$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Arithmetic error.

✗ **(C)** Omits factor of 2.

✗ **(D)** Forgets the length 10.

Question 60

★★★★★ VERY IMPORTANT

Correct Answer: (C) net change in position (displacement)

STEP-BY-STEP SOLUTION

By the Fundamental Theorem of Calculus, $\int_a^b v(t)\,dt = s(b) - s(a)$, the net displacement.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Acceleration integral would be $\int a(t)\,dt$.

✗ **(B)** Total distance requires integrating speed $|v(t)|$.

✗ **(D)** Average speed is $(1/(b-a)) \int |v|\,dt$.

Question 61

★★★★★ VERY IMPORTANT

Correct Answer: (C) velocity and acceleration have the same sign

STEP-BY-STEP SOLUTION

Speed is $|v|$. Its derivative is $d/dt\ |v| = v \cdot a / |v|$. Thus speed increases when v and a have the same sign.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** If $v < 0$, $a > 0$ would mean slowing down.

✗ **(B)** A particle can speed up while moving left (both negative).

✗ **(D)** Zero acceleration means constant velocity.

Question 62

★★★ MANDATORY

Correct Answer: (C) 3/5

STEP-BY-STEP SOLUTION

Form 0/0. Applying L'Hospital: $\lim [3 \sec^2(3x)] / [5 \cos(5x)] = 3(1) / [5(1)] = 3/5$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Inverts the ratio.

✗ **(B)** Assumes trig functions cancel to 1.

✗ **(D)** Both numerator and denominator approach 0.

Question 63

★★ IMPORTANT

Correct Answer: (C) the definition of the differential approximation

STEP-BY-STEP SOLUTION

This is the standard differential approximation $\Delta y \approx dy = f'(a)\ \Delta x$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** MVT guarantees an exact slope at an unknown c.

✗ **(B)** L'H evaluates indeterminate limits.

✗ **(D)** IVT guarantees roots/values.

Question 64

★★★ MANDATORY

Correct Answer: (D) 1/2 in/s

STEP-BY-STEP SOLUTION

$V = (4/3)\pi r^3 \Rightarrow dV/dt = 4\pi r^2 (dr/dt)$. $8\pi = 4\pi(4)(dr/dt) = 16\pi (dr/dt) \Rightarrow dr/dt = 8\pi / 16\pi = 1/2$ in/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses r^3 instead of r^2.

✗ **(B)** Multiplies instead of dividing.

✗ **(C)** Forgets $r^2 = 4$.

Question 65

★★★ MANDATORY

Correct Answer: (A) At t = 2 hours the temperature is dropping at 3°F per hour

STEP-BY-STEP SOLUTION

$T'(2)$ is the instantaneous rate of change. The negative value indicates temperature is dropping at 3°F/hr.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Confuses rate T' with temperature T.

✗ **(C)** Negative derivative indicates decrease.

✗ **(D)** Instantaneous rate, not an average over 2 hours.

Question 66

★★★ MANDATORY

Correct Answer: (C) 1/4

STEP-BY-STEP SOLUTION

Form 0/0. L'H 1: $(-\cos x) / (-2 \sin 2x)$ (still 0/0). L'H 2: $(\sin x) / (-4 \cos 2x) \to 1 / [-4(-1)] = 1/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The limit is finite.

✗ **(B)** Omits the chain rule factor of 2 in second derivative.

✗ **(D)** Stops prematurely.

Question 67

★★★★★ VERY IMPORTANT

Correct Answer: (B) 10

STEP-BY-STEP SOLUTION

v changes sign at t = 4. Distance = $\int_0^4 (4 - t)\,dt + \int_4^6 (t - 4)\,dt = [4t - t^2/2]_0^4 + [t^2/2 - 4t]_4^6 = 8 + 2 = 10$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is net displacement (8 − 2 = 6).

✗ **(C)** Only calculates the forward segment.

✗ **(D)** Adds an incorrect sign.

Question 68

★★★★★ VERY IMPORTANT

Correct Answer: (B) 20/3 ft/s

STEP-BY-STEP SOLUTION

Let x be distance from post and s be shadow length. 6/s = 15/(x+s) ⇒ 9s = 6x ⇒ s = (2/3)x. Tip position is x + s = (5/3)x. d/dt(x+s) = (5/3)(dx/dt) = (5/3)(4) = 20/3 ft/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Arithmetic error.

✗ **(C)** This is ds/dt, the rate shadow length grows.

✗ **(D)** This is just the person's speed.

Question 69

★★★★★ VERY IMPORTANT

Correct Answer: (B) 0.096 rad/s

STEP-BY-STEP SOLUTION

tan θ = y/3000. When y = 4000, hypotenuse = 5000, so cos θ = 3/5. $\sec^2\theta\,(d\theta/dt) = (1/3000)(dy/dt) \Rightarrow d\theta/dt = \cos^2\theta\,(dy/dt)/3000 = (9/25)(800)/3000 = 7200 / 75000 = 0.096$ rad/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses sin θ instead of cos θ.

✗ **(C)** Omits a factor of 2.

✗ **(D)** Forgets to square sec θ.

Question 70

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1, and it is an overestimate

STEP-BY-STEP SOLUTION

$f(0) = 1$, $f'(0) = -\sin(0) = 0$. $L(x) = 1 + 0(x) = 1$. Since $f''(x) = -\cos x < 0$ near 0, the curve is concave down, so $L(0.1) = 1$ is an overestimate.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Tangents are linear approximations, not exact values.

✗ **(B)** Concave down functions produce overestimates, not underestimates.

✗ **(C)** Uses an incorrect slope.

Question 71

★★★ MANDATORY

Correct Answer: (B) −1 cm²/s

STEP-BY-STEP SOLUTION

$A = lw \Rightarrow dA/dt = l(dw/dt) + w(dl/dt) = 8(-2) + 5(3) = -16 + 15 = -1$ cm²/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error on dw/dt.

✗ **(C)** Adds both positive rates.

✗ **(D)** Subtracts the product of rates.

Question 72

★★★★★ VERY IMPORTANT

Correct Answer: (A) (0, 1) and (1.5, 2)

STEP-BY-STEP SOLUTION

$v(t) = 6(t - 1)(t - 2)$, $a(t) = 6(2t - 3)$. Speed decreases when v and a have opposite signs: on (0, 1) ($v>0$, $a<0$) and on (1.5, 2) ($v<0$, $a>0$).

WHY THE OTHER OPTIONS FAIL

✗ **(B)** On (1, 1.5) speed is increasing because both are negative.

✗ **(C)** Ignores the velocity sign change at $t = 1$.

✗ **(D)** These are intervals where speed is increasing.

Question 73

★★★ MANDATORY

Correct Answer: (B) 0

STEP-BY-STEP SOLUTION

Form ∞/∞. Applying L'Hospital's Rule 3 times: ***lim*** 6 / (8e^(2x)) = 0.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Exponential strictly dominates polynomial.

✗ **(C)** Evaluates only numerator derivative.

✗ **(D)** Forgets that exponential remains in denominator.

Question 74

★★★ MANDATORY

Correct Answer: (C) 3/4

STEP-BY-STEP SOLUTION

Form 0/0. Applying L'Hospital: d/dx [√(4+3x)] / 1 = 3 / [2√(4+3x)]. At x = 0: 3 / [2(2)] = 3/4.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Forgets factor of 2 in denominator of derivative.

✗ **(B)** Assumes numerator vanishes completely.

✗ **(D)** Uses 1/4 from the square root.

Question 75

★★★ MANDATORY

Correct Answer: (A) 0.02 (2%)

STEP-BY-STEP SOLUTION

$A = s^2 \Rightarrow dA = 2s\,ds$. Relative error $dA/A = (2s\,ds) / s^2 = 2(ds/s) = 2(0.1 / 10) = 0.02$ (2%).

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Takes ds directly as relative error.

✗ **(C)** Squares the error factor.

✗ **(D)** Forgets the factor of 2 from the exponent.

Question 76

★★★ MANDATORY

Correct Answer: (B) 40

STEP-BY-STEP SOLUTION

Profit $P(x) = R(x) - C(x)$. $P'(x) = R'(x) - C'(x) = (50 - x) - 10 = 40 - x = 0 \Rightarrow x = 40$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Maximizes revenue, not profit.

✗ **(C)** Arithmetic error.

✗ **(D)** Uses fixed cost.

Question 77

★★★★★ VERY IMPORTANT

Correct Answer: (C) 4

STEP-BY-STEP SOLUTION

$v = 0$ at $t = 0.5$ and $t = 1.5$. Distance = $\int_0^{0.5} \pi \cos(\pi t)\, dt + \int_{0.5}^{1.5} -\pi \cos(\pi t)\, dt + \int_{1.5}^{2} \pi \cos(\pi t)\, dt = 1 + 2 + 1 = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is net displacement.

✗ **(B)** Only counts forward motion.

✗ **(D)** Multiplies by π incorrectly.

Question 78

★★★ MANDATORY

Correct Answer: (D) 4.5π m³/min

STEP-BY-STEP SOLUTION

$V = \pi R^2 h = 9\pi h$. $dV/dt = 9\pi (dh/dt) = 9\pi (-0.5) = -4.5\pi$ m³/min. Rate of decrease is 4.5π m³/min.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits radius.

✗ **(B)** Uses R instead of R^2.

✗ **(C)** Forgets to multiply by 0.5.

Question 79

★★★★★ VERY IMPORTANT

Correct Answer: (D) −1/3

STEP-BY-STEP SOLUTION

Form 0/0. L'H: $(1 - \sec^2 x) / (3x^2) = -\tan^2 x / (3x^2) = -(1/3)(\tan x / x)^2 \to -1/3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error using $\sec^2 x - 1$.

✗ **(B)** Premature substitution.

✗ **(C)** Omits the factor of 3 in denominator.

Question 80

★★★★★ VERY IMPORTANT

Correct Answer: (B) Prices are increasing, but the rate of inflation is slowing down

STEP-BY-STEP SOLUTION

$P' > 0$ means prices are increasing. $P'' < 0$ means the derivative P' is decreasing, so inflation (rate of price increase) is slowing down.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $P'' < 0$ means rate is decelerating, not accelerating.

✗ **(C)** $P' \neq 0$.

✗ **(D)** P' is positive, so prices are rising.

Question 81

★★★ MANDATORY

Correct Answer: (D) 3 + 1/27 ≈ 3.037

STEP-BY-STEP SOLUTION

$f(27) = 3$. $f'(x) = (1/3)x^{(-2/3)} \Rightarrow f'(27) = 1 / [3(9)] = 1/27$. $L(28) = 3 + (1/27)(1) = 3 + 1/27$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Arbitrary rounding.

✗ **(B)** Squares 27 incorrectly.

✗ **(C)** Omits the factor 1/3 from the power rule.

Question 82

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2/(3π) ft/min

STEP-BY-STEP SOLUTION

$r/h = 5/10 = 1/2 \Rightarrow r = h/2$. $V = (1/3)\pi(h/2)^2 h = (\pi/12)h^3$. $dV/dt = (\pi/4)h^2 (dh/dt) \Rightarrow 6 = (\pi/4)(36)(dh/dt) = 9\pi (dh/dt) \Rightarrow dh/dt = 6/(9\pi) = 2/(3\pi)$ ft/min.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Forgets the radius-height relationship.

✗ **(B)** Miscalculates derivative of h^3.

✗ **(D)** Inverts the fraction.

Question 83

★★★ MANDATORY

Correct Answer: (A) t = 1

STEP-BY-STEP SOLUTION

$v(t) = 4t^3 - 12t$, $a(t) = 12t^2 - 12$. $12t^2 - 12 = 0 \Rightarrow t^2 = 1 \Rightarrow t = 1$ for $t > 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Arithmetic error.

✗ **(C)** Divides incorrectly.

✗ **(D)** This is where velocity is zero.

Question 84

★★★★★ VERY IMPORTANT

Correct Answer: (C) 4

STEP-BY-STEP SOLUTION

Form 0/0. L'H: $[4 \sin 4x] / [\sin 2x + 2x \cos 2x]$ (0/0). L'H again: $[16 \cos 4x] / [2 \cos 2x + 2 \cos 2x - 4x \sin 2x] \rightarrow 16 / (2 + 2 - 0) = 16 / 4 = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Takes numerator as 0.

✗ **(B)** Forgets to scale by 4^2 in numerator.

✗ **(D)** Omits denominator derivative terms.

Question 85

★★★ MANDATORY

Correct Answer: (D) 3.5 °C/min

STEP-BY-STEP SOLUTION

$T'(t) = 70(-0.05)e^{(-0.05t)} = -3.5\ e^{(-0.05t)}$. At $t = 0$, $T'(0) = -3.5$ °C/min. The cooling rate is 3.5 °C/min.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Only uses the decay constant.

✗ **(B)** This is the room ambient temperature.

✗ **(C)** Confuses initial temperature difference with rate.

Question 86

★★★★★ VERY IMPORTANT

Correct Answer: (D) 25 knots

STEP-BY-STEP SOLUTION

$s^2 = x^2 + y^2 = (20t)^2 + (15t)^2 = 400t^2 + 225t^2 = 625t^2 \Rightarrow s = 25t$. Thus $ds/dt = 25$ knots at all $t > 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Directly adds the two speeds.

✗ **(B)** Averages the speeds.

✗ **(C)** Multiplies by the time elapsed (2 hours).

Question 87

★★★★★ VERY IMPORTANT

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

Rewrite as $(\ln x) / (\csc x)$, form $-\infty/\infty$. L'H: $(1/x) / (-\csc x \cot x) = -(\sin x \tan x)/x = -[\sin x / x] \cdot \tan x \to -1 \cdot 0 = 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error.

✗ **(B)** Incorrect limit evaluation.

✗ **(C)** $\ln x$ approaches $-\infty$ but sine forces convergence to 0.

Question 88

★★★★★ VERY IMPORTANT

Correct Answer: (A) overestimate

STEP-BY-STEP SOLUTION

$f''(x) = -\sin x < 0$ on $(0, \pi)$. Since f is concave down, the tangent line lies above the curve, so $L(0.05)$ is an overestimate.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Well-defined at all real numbers.

✗ **(C)** Linear approximations are not exact for trigonometric functions.

✗ **(D)** Concave up yields underestimates.

Question 89

★★★ MANDATORY

Correct Answer: (B) 23

STEP-BY-STEP SOLUTION

Average acceleration $= [v(3) - v(1)] / (3 - 1) = [(54 - 9) - (2 - 3)] / 2 = [45 - (-1)] / 2 = 46 / 2 = 23$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Arithmetic error in evaluating v(3).

✗ **(C)** Uses instantaneous acceleration at t = 2.

✗ **(D)** Forgets to divide by the time interval $\Delta t = 2$.

Question 90

★★★★★ VERY IMPORTANT

Correct Answer: (D) 6 cm²/s

STEP-BY-STEP SOLUTION

$A = (\sqrt{3} / 4)\, s^2 \Rightarrow dA/dt = (\sqrt{3} / 2)\, s\, (ds/dt) = (\sqrt{3} / 2)(4)(\sqrt{3}) = (3/2)(4) = 6$ cm²/s.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits the 1/2 factor from differentiation.

✗ **(B)** Forgets to multiply by $ds/dt = \sqrt{3}$.

✗ **(C)** Uses $(\sqrt{3}/4)\, s\, ds/dt$.

Question 91: Particle Motion from a Velocity Graph

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Acceleration is the slope of the velocity graph: $a(t) = v'(t)$.
On the interval $(2, 4)$, the line connects $(2, 2)$ and $(4, -2)$, so the slope is $(-2 - 2)/(4 - 2) = -2$ m/s².
Therefore, **$a(2.5) = -2$ m/s²**.
On $(4, 6)$, the graph is horizontal with slope 0, so **$a(5) = 0$ m/s²**.

(b) The particle moves to the left when velocity is strictly negative: $v(t) < 0$.
By linear interpolation, $v(t) = 0$ at $t = 3$ and $t = 7$. The graph lies strictly below the horizontal axis on **$(3, 7)$**, so the particle moves left for **$3 < t < 7$**.

(c) Total distance is the integral of speed: $\int_0^8 |v(t)|\, dt$, which equals the total geometric area bounded by $|v(t)|$ and the t-axis:
- On $[0, 2]$: rectangle of area $2 \times 2 = 4$.
- On $[2, 3]$: triangle of area $(1/2)(1)(2) = 1$.
- On $[3, 4]$: triangle of area $(1/2)(1)(2) = 1$.
- On $[4, 6]$: rectangle of area $2 \times 2 = 4$.
- On $[6, 7]$: triangle of area $(1/2)(1)(2) = 1$.
- On $[7, 8]$: triangle of area $(1/2)(1)(2) = 1$.

Total distance $= 4 + 1 + 1 + 4 + 1 + 1 =$ **12 meters**.

(d) Position is given by $s(8) = s(0) + \int_0^8 v(t)\, dt$.
The signed area is $(4 + 1) - (1 + 4 + 1) + 1 = 5 - 6 + 1 = 0$.
Therefore, $s(8) = 3 + 0 =$ **3 meters**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct acceleration $a(2.5) = -2$ and $a(5) = 0$ with units m/s²	2 pts
(b) Correct interval $(3, 7)$ with justification that $v(t) < 0$	2 pts
(c) Correct setup of geometric areas totaling 12 meters	3 pts
(d) Correct signed net displacement 0 and final position $s(8) = 3$ meters	2 pts

Question 92: Related Rates: Sliding Ladder

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) By the Pythagorean Theorem: $\mathbf{x^2 + y^2 = 169}$.

(b) Differentiating implicitly with respect to time t: $2x(dx/dt) + 2y(dy/dt) = 0 \Rightarrow x(dx/dt) + y(dy/dt) = 0$.
When $x = 5$, $y = \sqrt{(169 - 25)} = 12$. Given $dx/dt = 2$:
$5(2) + 12(dy/dt) = 0 \Rightarrow 12(dy/dt) = -10 \Rightarrow$ **dy/dt = −5/6 ft/s**.

(c) Since $dy/dt < 0$, the top of the ladder is **sliding down at a speed of 5/6 ft/s**.
The area of the triangle is $A = (1/2)xy$. Differentiating gives:
$dA/dt = (1/2)[x(dy/dt) + y(dx/dt)] = (1/2)[5(-5/6) + 12(2)] = (1/2)[-25/6 + 24] = (1/2)[119/6] =$ **119/12 ft²/s**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct Pythagorean relationship $x^2 + y^2 = 169$	2 pts
(b) Correct implicit differentiation with respect to time	2 pts
(b) Correct substitution yielding $dy/dt = -5/6$ ft/s	2 pts
(c) Identifies downward motion and correctly computes $dA/dt = 119/12$ ft²/s	3 pts

Question 93: Linear Approximation and Concavity

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $f(9) = \sqrt{9} = 3$.
$f'(x) = (1/2)x^{(-1/2)} = 1/(2\sqrt{x})$, so $f'(9) = 1/(2 \cdot 3) = 1/6$.
The linearization is **$L(x) = 3 + (1/6)(x - 9)$**.

(b) Evaluating at $x = 9.3$:
$L(9.3) = 3 + (1/6)(9.3 - 9) = 3 + (1/6)(0.3) = 3 + 0.05 =$ **3.05**.

(c) $f''(x) = d/dx\,[(1/2)x^{(-1/2)}] = -(1/4)x^{(-3/2)} = -1 / (4x^{(3/2)})$.
For all $x > 0$, $f''(x) < 0$, which means f is strictly concave down.
Because the graph of f lies entirely below its tangent line, the linear approximation $L(9.3) = 3.05$ is an **overestimate**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct derivative value $f'(9) = 1/6$ and formula for L(x)	3 pts
(b) Correct evaluation yielding 3.05	3 pts
(c) Computes $f''(x) < 0$ and justifies that concave down produces an overestimate	3 pts

Question 94: Tabular Analysis: Interpretation and Tangent Line

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $f'(2) = 3$ means that at time $x = 2$ hours, the temperature of the oven is **increasing at an instantaneous rate of 3 °C per hour**.

(b) At $x = 2$, $f(2) = 8$ and $f'(2) = 3$. The tangent line is $y - 8 = 3(x - 2) \Rightarrow L(x) = 8 + 3(x - 2)$.
$L(2.2) = 8 + 3(0.2) = 8 + 0.6 =$ **8.6 °C**.

(c) Since $f''(x) < 0$ on $[1, 4]$, the function is concave down on this interval, meaning the graph lies below its tangent line.
Therefore, the approximation 8.6 is an **overestimate**.

(d) Average rate of change $= [f(4) - f(1)] / (4 - 1) = (11 - 5) / 3 = 6 / 3 =$ **2 °C per hour**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Complete contextual interpretation with correct rate units (°C/hour)	2 pts
(b) Tangent line equation and correct value 8.6	3 pts
(c) Overestimate with justification from concavity ($f'' < 0$)	2 pts
(d) Correct difference quotient setup and evaluation 2 °C/hour	2 pts

Question 95: Related Rates: Expanding Spherical Balloon

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Differentiating volume with respect to time:
$dV/dt = 4\pi r^2 (dr/dt)$. Given $dV/dt = 8\pi$:
$8\pi = 4\pi r^2 (dr/dt) \Rightarrow$ **$dr/dt = 2 / r^2$**.

(b) When r = 2 inches:
$dr/dt = 2 / (2^2) = 2 / 4 =$ **1/2 inch per second**.

(c) Surface area is $A = 4\pi r^2$. Differentiating with respect to t:
$dA/dt = 8\pi r (dr/dt)$.
Substituting r = 2 and dr/dt = 1/2:
$dA/dt = 8\pi(2)(1/2) =$ **8π square inches per second**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Differentiates V with respect to t and solves for $dr/dt = 2/r^2$	3 pts
(b) Evaluates dr/dt = 1/2 in/s with correct units	3 pts
(c) Differentiates A and evaluates $dA/dt = 8\pi$ in²/s	3 pts

Question 96: Straight-Line Motion from an Algebraic Formula

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $v(t) = s'(t) = \mathbf{3t^2 - 12t + 9} = 3(t - 1)(t - 3)$.
$a(t) = v'(t) = \mathbf{6t - 12} = 6(t - 2)$.

(b) The particle is at rest when velocity equals zero: $3(t - 1)(t - 3) = 0 \Rightarrow$ **t = 1 second and t = 3 seconds**.

(c) The particle changes direction at t = 1 and t = 3. We evaluate position at the endpoints and turning points:

- $s(0) = 0$
- $s(1) = 1 - 6 + 9 = 4$
- $s(3) = 27 - 54 + 27 = 0$
- $s(4) = 64 - 96 + 36 = 4$

Total distance $= |s(1) - s(0)| + |s(3) - s(1)| + |s(4) - s(3)| = |4 - 0| + |0 - 4| + |4 - 0| = 4 + 4 + 4 =$ **12 meters**.

(d) At t = 2.5:
$v(2.5) = 3(2.5 - 1)(2.5 - 3) = 3(1.5)(-0.5) = -2.25 < 0$.
$a(2.5) = 6(2.5 - 2) = 6(0.5) = 3 > 0$.
Because velocity and acceleration have opposite signs, the speed of the particle is **decreasing**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct expressions for velocity v(t) and acceleration a(t)	2 pts
(b) Solves v(t) = 0 to get t = 1 and t = 3	2 pts
(c) Evaluates positions at all turning points and calculates distance = 12 m	3 pts
(d) Evaluates signs of v and a at t = 2.5 and concludes speed is decreasing	2 pts

Question 97: Indeterminate Forms and L'Hospital's Rule

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Direct substitution gives $(e^0 - 1) / \sin(0) = 0/0$, an indeterminate quotient.
Applying L'Hospital's Rule:
$\lim_{x \to 0} [3e^{(3x)}] / [2\cos(2x)] = [3(1)] / [2(1)] = \mathbf{3/2}$.

(b) As $x \to \infty$, $\ln x \to \infty$ and $\sqrt{x} \to \infty$, giving the indeterminate form ∞/∞.
Applying L'Hospital's Rule:
$\lim_{x \to \infty} (1/x) / [1 / (2\sqrt{x})] = \lim_{x \to \infty} (2\sqrt{x}) / x = \lim_{x \to \infty} 2 / \sqrt{x} = \mathbf{0}$.

(c) As $x \to 0^+$, this is the indeterminate product form $0 \cdot (-\infty)$.
Rewriting as a quotient: $\lim_{x \to 0^+} (\ln x) / (1/x)$, which is of form $-\infty/\infty$.
Applying L'Hospital's Rule:
$\lim_{x \to 0^+} (1/x) / (-1/x^2) = \lim_{x \to 0^+} (-x^2/x) = \lim_{x \to 0^+} (-x) = \mathbf{0}$.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) States form 0/0, differentiates numerator and denominator, evaluates 3/2	**3 pts**
(b) States form ∞/∞, differentiates and evaluates 0	**3 pts**
(c) Rewrites product as quotient of type $-\infty/\infty$ and evaluates limit as 0	**3 pts**

Question 98: Analysis of Motion from a Position Graph

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) On [0, 2], the line segment connects (0, 0) and (2, 4), so the average velocity is (4 − 0)/(2 − 0) = **2 m/s**. On (4, 6), the line segment connects (4, 4) and (6, 1). The instantaneous velocity is the slope: v(5) = (1 − 4)/(6 − 4) = −3/2 = **−1.5 m/s**.

(b) The particle is at rest when velocity v(t) = s′(t) = 0. The position graph is horizontal on **[2, 4]**, so the particle is at rest for **2 ≤ t ≤ 4**.

(c) The particle moves in the negative direction when velocity is negative: s′(t) < 0.
The position graph has a negative slope on **(4, 6]**.

(d) Distance traveled:
• From t = 0 to t = 2: moves from 0 to 4 (distance = 4 m).
• From t = 2 to t = 4: stationary (distance = 0 m).
• From t = 4 to t = 6: moves from 4 to 1 (distance = $|1 - 4|$ = 3 m).
Total distance = 4 + 0 + 3 = **7 meters**.
Displacement = s(6) − s(0) = 1 − 0 = **1 meter**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Computes slopes 2 m/s and −1.5 m/s with units	2 pts
(b) Identifies interval [2, 4] where slope is 0	2 pts
(c) Identifies interval (4, 6] where slope is negative	2 pts
(d) Computes distance = 7 m and displacement = 1 m	3 pts

Question 99: Applied Rate of Change from Tabular Data

★★★ MANDATORY

COMPLETE MODEL SOLUTION

(a) Using the values at t = 5 and t = 15:
V′(10) ≈ [V(15) − V(5)] / (15 − 5) = (66 − 95) / 10 = −29 / 10 = **−2.9 gallons per minute**.

(b) At time t = 10 minutes, the volume of water in the container is **decreasing at an estimated rate of 2.9 gallons per minute**.

(c) Average rate of change on [0, 15] = [V(15) − V(0)] / (15 − 0) = (66 − 120) / 15 = −54 / 15 = **−3.6 gallons per minute**.

(d) On [0, 5]: rate = (95 − 120) / 5 = −25 / 5 = −5 gal/min (loss of 5 gal/min).
On [10, 15]: rate = (66 − 78) / 5 = −12 / 5 = −2.4 gal/min (loss of 2.4 gal/min).
Since $|-5| > |-2.4|$, the water is leaking **faster during the first 5 minutes**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Symmetric difference quotient setup and evaluation −2.9 gal/min with units	3 pts
(b) Correct contextual interpretation noting rate of decrease	2 pts
(c) Average rate of change −3.6 gal/min with units	2 pts
(d) Compares rate magnitudes and concludes faster during the first 5 minutes	2 pts

Question 100: Related Rates: Inverted Conical Reservoir

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) By similar triangles in the cross-section of the cone:
$r / h = 3 / 12 = 1 / 4 \Rightarrow$ **$r = h / 4$**.
The volume is $V = (1/3)\pi r^2 h = (1/3)\pi(h/4)^2 h = (1/3)\pi(h^2/16)h =$ **$(\pi / 48) h^3$**.

(b) Differentiating volume with respect to time:
$dV/dt = (\pi / 48)(3h^2)(dh/dt) = (\pi / 16)\, h^2\, (dh/dt)$.
Given $dV/dt = 10$, when $h = 4$:
$10 = (\pi / 16)(4^2)(dh/dt) = (\pi / 16)(16)(dh/dt) = \pi\, (dh/dt) \Rightarrow$ **$dh/dt = 10 / \pi$ ft/min**.

(c) From $dV/dt = 10 = (\pi / 16)\, h^2\, (dh/dt)$, we have $dh/dt = 160 / (\pi\, h^2)$.
Since dh/dt is inversely proportional to h^2, as h increases, dh/dt decreases.
At $h = 4$, $dh/dt = 10/\pi \approx 3.18$ ft/min; at $h = 8$, $dh/dt = 160 / (64\pi) = 2.5/\pi \approx 0.80$ ft/min.
Therefore, the water level is rising **faster when h = 4 feet**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Uses similar triangles to find $r = h/4$ and derives $V = (\pi/48)h^3$	3 pts
(b) Differentiates with respect to t and finds $dh/dt = 10/\pi$ ft/min	4 pts
(c) Justifies that rate is inversely proportional to h^2 and faster at $h = 4$	2 pts

Essential Unit 4 Formulas and Theorems

- **Meaning of the Derivative:** f′(a) represents the instantaneous rate of change of f with respect to x at x = a. Units = (units of f)/(units of x).
- **Straight-Line Motion:** Velocity v(t) = s′(t), Acceleration a(t) = v′(t) = s″(t). Speed = |v(t)|. Particle at rest when v(t) = 0.
- **Speeding Up vs Slowing Down:** Speed is increasing when v(t) and a(t) have the same sign; decreasing when they have opposite signs.
- **Displacement vs Distance:** Displacement = ∫_a^b v(t) dt = s(b) − s(a). Total distance = ∫_a^b |v(t)| dt.
- **Linearization:** Tangent line approximation L(x) = f(a) + f′(a)(x − a). If f″(x) > 0, L(x) is an underestimate; if f″(x) < 0, L(x) is an overestimate.
- **Differential Approximation:** dy = f′(x) dx, where Δy ≈ dy for small dx. Relative error = dy / y.
- **Related Rates Strategy:** 1) Identify given rates and desired rate; 2) Relate variables with an equation; 3) Differentiate implicitly with respect to t; 4) Substitute numerical values.
- **L'Hospital's Rule:** For indeterminate limits of form 0/0 or ±∞/±∞, lim [f(x)/g(x)] = lim [f′(x)/g′(x)], provided the limit exists or is infinite.

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